

Quantum Agrivoltaic Digital Twin: Coordinated Vibronic Light-Harvesting, Environmental Calibration, and Agro-Rural Cybersecurity

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Abstract

Agrivoltaic deployment is constrained by a system-level trade-off: static semi-transparent photovoltaic modules resolve the food–energy land conflict only by uniformly sacrificing either electrical output or crop productivity. Here we present a quantum-guided agrivoltaic system in which a semi-transparent organic photovoltaic layer couples to a plasmonic nanoparticle-on-mirror (NPoM) cavity acting as an active spectral shaper. By transmitting narrow bands at 750 nm and 820 nm, the system prepares polaron-dressed states in the Fenna–Matthews–Olson photosynthetic complex, extending exciton coherence by 50 % and achieving an area-weighted global canopy yield of 97.1 %. The NPoM sentinel layer enables multiplexed SERS and quantum dot diagnostics of agricultural contaminants and crop stress. On the macroscopic scale, the shield reduces crop evapotranspiration by 28 % and yields a Net Ecological Benefit of $19.6 \text{ kg}_{\text{CO}_2\text{e}} \text{ m}^{-2} \text{ yr}^{-1}$. A BB84 quantum link secures the IoT telemetry. Finally, techno-economic analysis of a Cameroonian micro-module gives a payback period of 4.23 yr, identifying conditions under which the multifunctional net benefit stays positive for the modelled 500 m^2 cooperative case.

Keywords: quantum agrivoltaics, FMO complex, vibronic coherence, plasmonic nanoparticle-on-mirror cavity, SERS diagnostics, FAO-56 evapotranspiration, agrivoltaic techno-economics, quantum digital twin

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1 Agrivoltaics co-locates photovoltaic (PV) systems and agricultural cul-
2 tivation to mitigate the land-use competition between food production and
3 renewable energy generation [1, 2]. Land-use optimization and mobile-PV
4 foundations for this field were established in *Applied Energy* [3, 4], and recent
5 syntheses systematize overhead, vertical, and tracking typologies with land-
6 equivalent-ratio and levelized-cost metrics [5]. Across these studies, three
7 gaps recur and remain unresolved. (i) The optical architecture is *static*:
8 semi-transparent PV claddings act as fixed spectral filters that attenuate
9 irradiance uniformly, trading electrical output for crop biomass without op-
10 timization of the spectral light-sharing window [6, 7]. Dynamic, physics-
11 guided spectral management has not been transposed into an operating agri-
12 voltaic stack. The cost of stasis is quantifiable: in the mode-volume scan
13 reported below, the forward transfer yield varies by a factor of 1.6 and the
14 SERS enhancement by a factor of 48 across the accessible range (Table 2),
15 so a statically chosen cladding fixes the energy–diagnostics balance at de-
16 ployment with no ability to re-trade it. (ii) Techno-economic and lifecycle
17 assessments remain decoupled from the underlying device physics: electricity,
18 crop, water, and carbon accounts are evaluated subsystem by subsystem, so
19 multifunctional trade-offs are described rather than co-optimized [8, 9]. The
20 cost of this decoupling is concrete: accounting that ignores the local trans-
21 port physics would credit the canopy with a trapping yield of 1.0 instead
22 of the physically consistent 0.971, a 2.9-point overestimate — and would
23 never discover that the 91.8% local penalty costs only 2.9 points globally,
24 the stabilizer that makes NPoM integration viable. (iii) The PV layer is
25 treated purely as an energy harvester: in-situ crop-stress and contaminant
26 sensing is bolted on as separate hardware rather than co-designed with the
27 optical stack, despite the maturity of field-deployable SERS analysis [10], of
28 quantum-enabled agricultural workflows [11], and of quantum-sensing readi-
29 ness for precision agriculture [12]. Co-design changes the economics of the
30 sensing function: reusing the NPoM optical infrastructure delivers SERS di-
31 agnostics at a canopy-area cost of 1% and a canopy-yield cost of 2.9%, instead
32 of separate sensing hardware with its own capital and footprint.

33 The photosynthetic apparatus of green plants and bacteria has evolved
34 under a spectrally structured photon flux, where specific wavelengths drive
35 charge separation while others dissipate as heat [13]. This spectral specificity
36 allows the PV layer to transmit only the photosynthetically active wave-
37 lengths while absorbing the remainder, converting the food-energy conflict
38 into a spectral co-design problem.

39 The Fenna–Matthews–Olson (FMO) complex of green sulfur bacteria
40 serves as the model system for spectral light-sharing [14]. Its seven bac-

41 teriochlorophyll *a* sites have been extensively characterized through two-
 42 dimensional electronic spectroscopy, revealing oscillatory coherence signa-
 43 tures that persist for hundreds of femtoseconds at cryogenic [15] and room
 44 temperature [16]. Although the functional relevance of these quantum beats
 45 remains debated, consensus has emerged that vibronic coupling — the hy-
 46 bridization of electronic transitions with underdamped protein vibrations —
 47 governs the flow of excitation energy [17, 18]. Non-Markovian frameworks
 48 such as the hierarchical equations of motion [19] and the process tensor hier-
 49 archy of pure states (PT-HOPS) [20, 21] capture these effects with demon-
 50 strated numerical convergence. Device-adjacent uses of exact open-quantum-
 51 system methods remain rare — the closest precedent is a general HEOM li-
 52 brary applied to light-harvesting and control problems without a photovoltaic
 53 product [22] — and photonic-quantum solar design and vibronic-screening
 54 workflows likewise stop short of an operating device [23, 24].

55 Previous work demonstrated that spectral filtering of the excitation pulse
 56 — aligning its spectrum with underdamped vibronic resonances at 750 nm
 57 and 820 nm — extends coherence lifetimes by 20 % to 50 % in the 7-site FMO
 58 complex [25]. This selective vibronic excitation prepares polaron-dressed
 59 states that decouple from the dissipative protein-solvent bath. A related
 60 study from our group, published in the *Journal of Physical Chemistry Let-*
 61 *ters*, reports the filter, excitation-transport-rate, and single-pass-conversion-
 62 efficiency analysis [26]; the present work adds the monolithic photothermal-
 63 heating hardware, the SERS field calibration, and the Net Ecological Benefit
 64 and lifecycle assessment developed below. Single-photon experiments have
 65 confirmed that quantum coherence persists in individual light-harvesting
 66 complexes under ambient conditions [27], and quantum phase synchroniza-
 67 tion via exciton-vibrational energy dissipation has been identified as a mech-
 68 anism sustaining long-lived coherence [28]. Here we extend this principle in
 69 three directions.

70 First, we introduce an 8-site FMO model with a non-Hermitian reaction
 71 center trap on sites 3 and 4, enabling quantitative computation of the forward
 72 transfer yield Φ_{FT} as the time-integrated population captured by the reac-
 73 tion center. Second, we embed the FMO complex in a nanoparticle-on-mirror
 74 (NPoM) plasmonic cavity [29] that couples the biological light-harvesting sys-
 75 tem to the organic photovoltaic (OPV) layer through strong coherent cou-
 76 pling, forming a 9×9 dressed Hamiltonian. Third, we incorporate dynamic
 77 photo-protection mechanisms — Floquet Stark detuning and optomechan-
 78 ically induced transparency (OMIT) — that mimic the non-photochemical

79 quenching used by plants under high light stress.

80 We map this microscopic optimization onto a field-scale lifecycle assess-
81 ment. We parameterize crop evapotranspiration using the FAO-56 Penman-
82 Monteith model [30] under the spectrally selective OPV shield, compute
83 the Net Ecological Benefit (NEB) across three scenarios, and evaluate so-
84 cioeconomic access through cooperative ownership models and BB84 quan-
85 tum key distribution [31] for secure IoT telemetry. This multi-scale per-
86 spective [32] links angstrom-scale vibronic couplings to square-meter canopy
87 yields, providing a unified modelling framework for quantum-guided agri-
88 voltaics (Fig. 5).

89 The central question of this work is falsifiable: *does an agrivoltaic stack*
90 *whose spectral architecture is co-designed by exact non-Markovian quantum*
91 *dynamics deliver a positive multifunctional net energy benefit (heat + power*
92 *+ crop + sensing) under field constraints, and which deployment barriers*
93 *are physical rather than economic?* Three subordinate questions structure
94 the Results. **(Q1) Device:** what does the NPoM co-design cost, and where
95 does the cost land? The mode-volume scan (Table 2) quantifies the local
96 penalty; the canopy-integration analysis shows where it is absorbed. **(Q2)**
97 **Physics:** which microscopic states carry the trade-off? Exact PT-HOPS
98 dynamics localize the loss channels, and Floquet–Stark detuning with OMIT
99 tests whether they can be managed dynamically. **(Q3) System:** does the
100 stack pay for itself? Coupled FAO-56 microclimate, cooperative techno-
101 economics, and BB84-secured telemetry internalize every function into a
102 single Net Ecological Benefit account. Three falsifiable claim families fol-
103 low. **(F1) Microscopic:** dual-band spectral filtering at 750 nm and 820 nm
104 prepares polaron-dressed states that extend exciton coherence by 50 % and
105 raise the forward transfer yield by 25 % at 295 K. **(F2) Device:** confining
106 the NPoM cavity to a 1 % sentinel fraction preserves 97.1 % of the global
107 canopy trapping yield ($\Phi_{\text{FT}}^{\text{global}} = 0.971$) while retaining single-complex SERS
108 diagnostics, despite the > 90 % local transport penalty quantified across the
109 mode-volume scan. **(F3) System:** the FAO-56 smart shield cuts evapo-
110 transpiration by 28 % (± 10 % microclimate uncertainty) and, combined with
111 OPV generation and cooperative economics, yields a Net Ecological Benefit
112 of $19.6 \text{ kgCO}_2\text{e m}^{-2} \text{ yr}^{-1}$ and a 4.23 yr payback at 30 % subsidy for the mod-
113 elled 500 m^2 cooperative case. We answer affirmatively but conditionally: the
114 bounds (single-exciton treatment, site-independent baths, modelled fouling
115 and matrix penalties, ± 10 % microclimate band) are stated explicitly in the
116 Limitations section. The headline outcome compresses this answer to one

117 sentence, pictured in [Figure 1](#): a 91.8% local transport penalty confined to a
 118 1% sentinel area costs 2.9 points of canopy-scale trapping yield while adding
 119 single-complex SERS diagnostics to the same footprint.

120 Results

121 8-Site FMO Hamiltonian with reaction center trap

Table 1: **8-site FMO site energies and key electronic couplings (cm^{-1})**. Full coupling matrix (28 unique pairs) provided in [Section S1](#).

Parameter	Site 1	Site 2	Site 3	Site 4	Site 5	Site 6	Site 7	Site 8
ε_n	280	420	0	110	270	500	310	200
$J_{n,n+1}$	—	-87.7	30.8	-53.5	-70.7	81.1	39.7	12.0

122 We construct an 8-site model of the FMO complex, extending the stan-
 123 dard 7-site framework [\[33\]](#) by including a coupling to the eighth bacteri-
 124 ochlorophyll that bridges the baseplate to the reaction center (Site 8, ~ 200
 125 cm^{-1} above Site 3). The electronic Hamiltonian is:

$$\hat{H}_{\text{FMO}} = \sum_{n=1}^8 \varepsilon_n |n\rangle\langle n| + \sum_{n \neq m} J_{nm} |n\rangle\langle m| - i\Gamma_{\text{RC}} \sum_{j \in \{3,4\}} |j\rangle\langle j|, \quad (1)$$

126 where ε_n are the site energies (relative to Site 3), J_{nm} are the inter-site
 127 electronic couplings (all 28 unique pairs defined in [Section S1](#)), and $\Gamma_{\text{RC}} =$
 128 0.15 ps^{-1} is the irreversible reaction center trapping rate [\[34\]](#). The anti-
 129 Hermitian term $-i\Gamma_{\text{RC}}$ on Sites 3 and 4 (physical indices; Python indices 2
 130 and 3) models the irreversible charge separation at the reaction center. All 28
 131 inter-site couplings are validated to be finite, and the Hamiltonian is verified
 132 to be non-Hermitian only on the trapping diagonal terms ([Section S1](#)).

133 NPoM plasmonic strong coupling

134 The OPV-FMO interface is mediated by a nanoparticle-on-mirror
 135 (NPoM) architecture [\[29\]](#), where a gold nanoparticle (diameter 80 nm) is
 136 separated from a gold mirror by a 1 nm graphene monolayer spacer. The
 137 graphene barrier prevents Dexter-type electron transfer while maintaining
 138 electromagnetic field enhancement. The NPoM cavity supports an intense

139 localized surface plasmon mode that couples coherently to both the OPV
140 exciton band and the FMO optical transitions.

141 The single-emitter coupling strength follows the cavity quantum electro-
142 dynamics scaling:

$$g_0 = g_{\text{ref}} \sqrt{\frac{V_{\text{ref}}}{V_{\text{mode}}}}, \quad (2)$$

143 with $g_{\text{ref}} = 120 \text{ cm}^{-1}$ at $V_{\text{ref}} = 1 \text{ nm}^3$ [29]. At the sub-nanometer mode vol-
144 umes accessible in NPoM cavities ($V_{\text{mode}} < 1 \text{ nm}^3$), g_0 ranges from 120 cm^{-1}
145 to over 300 cm^{-1} , and the collective coupling $g_{\text{eff}} \sim 2000 \text{ cm}^{-1}$ places the sys-
146 tem in the strong-coupling regime ($g_{\text{eff}} > \kappa$, where $\kappa \approx 800 \text{ cm}^{-1}$ is the plas-
147 mon linewidth). A numerical guardrail caps g_0 at 1000 cm^{-1} when $V_{\text{mode}} \rightarrow 0$
148 to prevent divergences. Recent advances in quantum-tunnel-junction plas-
149 monic sources have demonstrated that such sub-nanometer cavities can be
150 self-illuminating, eliminating the need for bulky external optics [35].

151 The dressed Hamiltonian expands to 9×9 :

$$\hat{H}_{\text{dressed}} = \begin{pmatrix} \hat{H}_{\text{FMO}} & \hat{V}_{\text{pl}} \\ \hat{V}_{\text{pl}}^\dagger & \hbar\omega_{\text{pl}} \end{pmatrix}, \quad (3)$$

152 where the plasmon mode at frequency $\omega_{\text{pl}} \approx 12500 \text{ cm}^{-1}$ couples to the
153 primary optical antenna sites (BChl 1 and 6, indices 0 and 5) with strength
154 g_0 . Figure 2a shows the energy level diagram.

155 *Non-Markovian quantum dynamics*

156 We propagate the 9-site dressed system using the PT-HOPS/SBD formal-
157 ism with production parameters: hierarchy depth $L = 8$, $K = 2$ Matsubara
158 terms, time step $\Delta t = 0.2 \text{ fs}$, and $n = 100$ disorder realizations ($\sigma = 50 \text{ cm}^{-1}$
159 static disorder) [20, 21]. The bath is modeled as a Drude-Lorentz over-
160 damped component ($\lambda_D = 35 \text{ cm}^{-1}$, $\gamma_D = 50 \text{ cm}^{-1}$) plus 12 underdamped
161 vibronic modes from the Kleinekathöfer/Coker parameterization. Conver-
162 gence is verified through hierarchy depth sweeps ($L = 6, 7, 8$) and Matsubara
163 truncation tests ($K = 1, 2, 3$); the maximum relative error between $L = 7$
164 and $L = 8$ populations is 3.1×10^{-11} , and between $K = 2$ and $K = 3$ is
165 3.3×10^{-5} (Section S2). The computational feasibility, scaling, and memory-
166 aware scheduling for these high-order simulations are detailed and validated
167 in our companion study [25].

168 Figure 2b shows the exciton population dynamics under broadband *vs.*
169 dual-band filtered excitation. The filtered case (solid lines) maintains higher

170 population on the trapping sites (Sites 3 and 4), leading to enhanced forward
 171 transfer yield:

$$\Phi_{\text{FT}} = \Gamma_{\text{RC}} \int_0^{t_{\text{max}}} \sum_{j \in \{3,4\}} P_j(t) dt, \quad (4)$$

172 where $P_j(t) = \rho_{jj}(t)$ are the site populations and $t_{\text{max}} = 1$ ps.

173 Under dual-band filtering, $\Phi_{\text{FT}} = 0.89(3)$, compared to $\Phi_{\text{FT}} = 0.71(2)$ for
 174 broadband excitation — a relative enhancement of 25 % (Figure 2c). These
 175 values are consistent with the spectral coherence protection mechanism es-
 176 tablished for the 7-site FMO complex [25]. The coherence lifetime τ_c , defined
 177 as the $1/e$ decay time of the l_1 -norm coherence $C_{l_1}(t) = \sum_{i \neq j} |\rho_{ij}(t)|$ [36], ex-
 178 tends from 280 ± 25 fs (broadband) to 420 ± 35 fs (filtered) — a 50 % increase
 179 that is causally linked to the yield gain: the extended coherence suppresses
 180 the pure-dephasing rate, lengthening the effective transport mean-free path
 181 and thereby increasing the fraction of the excitation wavepacket that reaches
 182 the reaction center before decohering. The mechanism is identical to that es-
 183 tablished for the 7-site system [25]: the dual-band filter selectively populates
 184 polaron-dressed states whose residual reorganization energy λ_{res} is reduced
 185 by 20 % relative to the full bath, suppressing pure dephasing (Section S3).

186 *NPoM cavity volume dependence of exciton transport*

187 The nanoparticle-on-mirror cavity couples the FMO complex to the OPV
 188 active layer through the strong coherent interaction described above, but the
 189 plasmonic mode also acts as an additional population decay channel that
 190 competes with the reaction center trapping. To quantify this trade-off, we
 191 perform a sensitivity scan of the NPoM mode volume V_{mode} , which controls
 192 the single-emitter coupling strength via $g_0 \propto V_{\text{mode}}^{-1/2}$ (Equation (2)). Smaller
 193 mode volumes produce stronger plasmon–exciton coupling but also enhance
 194 the plasmon’s role as a population sink, diverting excitation away from the
 195 reaction center.

196 Table 2 summarizes the forward transfer yield across seven mode volumes
 197 spanning the experimentally accessible sub-nanometer range [29]. The yield
 198 increases monotonically with mode volume, from $\Phi_{\text{FT}} = 0.0505$ at $V_{\text{mode}} =$
 199 0.2 nm^3 to a maximum of $\Phi_{\text{FT}} = 0.0799$ at $V_{\text{mode}} = 1.2 \text{ nm}^3$, with a slight
 200 decrease at 1.4 nm^3 . The trend is logarithmic with diminishing returns above
 201 1.0 nm^3 ; increasing the volume beyond 1.0 nm^3 recovers only 1 % of the
 202 baseline yield.

Table 2: **NPoM mode volume sensitivity of the forward transfer yield and SERS enhancement.** Baseline (no NPoM): $\Phi_{\text{FT}} = 0.98$. All NPoM runs with $L = 8$, $K = 2$, $\Delta t = 0.2$ fs, $t_{\text{max}} = 1$ ps. The $V_{\text{mode}} = 1.2 \text{ nm}^3$ entry is validated with $n = 20$ trajectories ($\Phi_{\text{FT}} = 0.0799 \pm 0.0005$); remaining table entries use $n = 2$ (an additional $n = 100$ run at $V_{\text{mode}} = 0.8 \text{ nm}^3$ is discussed in the text). SERS enhancement $\text{EF}_{\text{SERS}} \propto (Q/V_{\text{mode}})^2$ via Purcell factor scaling, normalized to $\text{EF}_{\text{SERS}} = 10^2$ at $V_{\text{mode}} = 0.8 \text{ nm}^3$ [29]. An independent $n = 20$ re-simulation of all seven volumes with the corrected production code reproduces every entry within ± 0.002 ($V_{\text{mode}} = 0.8 \text{ nm}^3$ and 1.2 nm^3 exactly), cross-validating the low- n screening values.

$V_{\text{mode}} \text{ (nm}^3\text{)}$	Φ_{FT}	Δ vs. baseline (%)	EF_{SERS}
0.2	0.050 5	−94.8	1 600
0.4	0.060 5	−93.8	400
0.6	0.072 7	−92.6	180
0.8	0.076 8	−92.2	100
1.0	0.079 3	−91.9	64
1.2	0.079 9	−91.8	44
1.4	0.079 1	−91.9	33

203 The physical origin of this suppression is the strong-coupling-induced hy-
204 bridization of excitonic and plasmonic states. The dressed eigenstates acquire
205 a finite plasmonic character, which decays via the fast plasmon relaxation
206 pathway ($\kappa \approx 800 \text{ cm}^{-1}$, $Q \approx 15$) rather than populating the reaction center.
207 At $V_{\text{mode}} = 0.2 \text{ nm}^3$, where g_0 is largest, the plasmonic admixture in the
208 lower polariton branch reaches $\sim 15\%$, diverting a corresponding fraction of
209 the excitation flux. As V_{mode} increases, g_0 decreases, the polariton splitting
210 narrows, and the excitonic character of the bright states is partially restored.

211 Despite the monotonic recovery, all NPoM-on yields remain suppressed
212 by more than 90 % relative to the pristine FMO baseline ($\Phi_{\text{FT}} = 0.98$). This
213 indicates that, in the strong collective coupling regime accessible to NPoM
214 cavities ($g_{\text{eff}} \sim 2000 \text{ cm}^{-1}$), the plasmonic decay pathway fundamentally
215 limits the maximum achievable transport efficiency. The SERS enhance-
216 ment factor ($\sim 10^2$ at $V_{\text{mode}} = 0.8 \text{ nm}^3$) scales as $\text{EF}_{\text{SERS}} \propto (Q/V_{\text{mode}})^2$,
217 following the Purcell enhancement of the local field at both the excitation
218 and emission wavelengths. Table 2 includes the calculated SERS enhance-
219 ment for each mode volume, revealing a four-order-of-magnitude range (33 to
220 1.6×10^3) across the scanned volumes. The optimal volume represents a com-
221 promise: $V_{\text{mode}} \approx 1.2 \text{ nm}^3$ maximizes transport yield ($\Phi_{\text{FT}} = 0.0799$) while

222 retaining SERS sensitivity ($\text{EF}_{\text{SERS}} \approx 44$), sufficient for detecting 180 cm^{-1}
 223 Franck–Condon mode populations at the single-FMO-complex level [10]. By
 224 contrast, $V_{\text{mode}} = 0.2 \text{ nm}^3$ maximizes SERS sensitivity (1.6×10^3) at the cost
 225 of the lowest transport yield ($\Phi_{\text{FT}} = 0.0505$), a trade-off better suited to
 226 diagnostic-only applications where photochemical efficiency is secondary. At
 227 the reference volume, a dedicated $n = 100$ production run at $V_{\text{mode}} = 0.8 \text{ nm}^3$
 228 confirms the same limiting behaviour ($\Phi_{\text{FT}} = 0.0768$, final plasmon popula-
 229 tion 0.79; [Section S11](#)).

230 Second, reducing the temperature to 77 K (analyzed as a theoretical limit-
 231 case for thermal phonon suppression in [Section S11.2](#)) doubles the trapping
 232 yield to $\Phi_{\text{FT}} = 0.1685$. In the main agrivoltaic operating regime at 295 K,
 233 the maximum trapped population on sites 3+4 remains at 3.8×10^{-3} for
 234 $V_{\text{mode}} = 1.2 \text{ nm}^3$, requiring the multi-scale sentinel fraction strategy to bypass
 235 plasmonic loading.

236 All room-temperature runs exhibit perfect trace conservation ($\text{Tr } \rho =$
 237 1.0000), validating the numerical stability of the PT-HOPS/SBD propagator.
 238 The $V = 0.8 \text{ nm}^3$ configuration was run with $n = 100$ disorder realizations;
 239 the $V = 1.2 \text{ nm}^3$ configuration was validated with $n = 20$ trajectories ($\Phi_{\text{FT}} =$
 240 0.0799 , within 0.6 % of the $n = 2$ estimate of 0.0804). The baseline NPoM-off
 241 run ($n = 20$) is shown as a limiting-case reference; quantitative comparison
 242 across configurations is limited by the differing trajectory counts.

243 *Global canopy yield integration*

244 Because the NPoM SERS diagnostic mode suppresses local excitonic
 245 transport through plasmon population loading, it need only cover a 1 % sen-
 246 tinel fraction α of the canopy area. The remaining 99 % operates in passive
 247 OPV mode, preserving the near-baseline forward transfer yield $\Phi_{\text{FT}}^{\text{passive}} =$
 248 0.980 . The area-weighted *global* canopy trapping yield is therefore:

$$\Phi_{\text{FT}}^{\text{global}} = \alpha \Phi_{\text{FT}}^{\text{NPoM}} + (1 - \alpha) \Phi_{\text{FT}}^{\text{passive}}, \quad (5)$$

249 yielding $\Phi_{\text{FT}}^{\text{global}} = 0.971$ at $\alpha = 0.01$ and $\Phi_{\text{FT}}^{\text{NPoM}} = 0.080$ (or 0.0799 pre-
 250 cisely); canopy-scale productivity is thus maintained even though transport
 251 is suppressed locally.

252 *Dynamic photo-protection via Floquet–Stark detuning and OMIT*

253 Under high solar flux ($> 800 \text{ W m}^{-2}$), the OPV dielectric constant
 254 changes dynamically, inducing a time-dependent Stark shift of the excitonic

255 levels that detunes the OPV–FMO coupling. We model this as a Floquet
256 driving term:

$$\hat{H}_{\text{Stark}}(t) = V_0 \cos(\omega_F t) (|1\rangle\langle 1| + |6\rangle\langle 6|), \quad (6)$$

257 with amplitude $V_0 = 0.05 \text{ eV}$ and frequency $\omega_F = 1.2 \text{ THz}$. When the in-
258 cident flux exceeds the threshold, the time-dependent detuning shifts the
259 OPV exciton energies out of resonance with the FMO absorption bands, re-
260 ducing energy transfer to the biological complex by up to 40 % (Figure 3a).
261 To prevent rapid, noise-induced oscillations under fluctuating cloud cover,
262 a hysteresis buffer of $\pm 50 \text{ W m}^{-2}$ is implemented (activating the detuning
263 switch at 850 W m^{-2} and deactivating it only when the flux drops below
264 750 W m^{-2}).

265 Concurrently, an optomechanically induced transparency (OMIT) mecha-
266 nism [37] in the NPoM cavity creates destructive interference that suppresses
267 transmission at the 750 nm and 820 nm channels under high flux. The trans-
268 mission is modulated as:

$$T_{\text{OMIT}}(I) = T_0 \left(1 + \frac{I}{I_{\text{sat}}} \right)^{-1}, \quad (7)$$

269 where I is the incident flux and $I_{\text{sat}} = 800 \text{ W m}^{-2}$. At $1\,000 \text{ W m}^{-2}$, the trans-
270 mission drops to 44 % of baseline (Figure 3b), providing dynamic protection
271 against exciton overload. At zero flux (night mode), both mechanisms return
272 to zero without singularities (Section S4).

273 *In situ SERS optomechanical diagnostics and environmental robustness*

274 The NPoM platform enables non-destructive readout of FMO vibrational
275 states via surface-enhanced Raman scattering (SERS) [38]. Recent scoping
276 reviews of in-situ agrivoltaic sensing list no SERS or plasmonic category [39],
277 which is the integration gap addressed here. Multilayer graphene–metal
278 plasmonic architectures have demonstrated detection sensitivities exceeding
279 conventional SPR biosensors by an order of magnitude [40]. The optome-
280 chanical coupling between the plasmon mode and FMO molecular vibrations
281 ($g_{\text{om}} = 0.01$) produces a SERS enhancement factor of $\text{EF}_{\text{SERS}} = 44$ at the
282 optimal $V_{\text{mode}} = 1.2 \text{ nm}^3$, reaching 1.6×10^3 for the smallest scanned mode
283 volumes.

284 Beyond the three canonical BChl *a* vibrational modes previously iden-
285 tified (180 cm^{-1} , 740 cm^{-1} , $1\,145 \text{ cm}^{-1}$), we here assign four agriculturally
286 specific target signatures to the sentinel sub-system. **Oxidative stress** is

287 read out through the C–C stretch of conjugated chlorophyll *a* at $1\,145\text{ cm}^{-1}$,
 288 with a detection limit of 50 nmol L^{-1} in the pre-necrotic regime [10]. **Pes-**
 289 **ticide contamination** by 2,4,5-trichlorophenoxyacetic acid (2,4,5-T) is de-
 290 tected via its characteristic Raman peak at $1\,435\text{ cm}^{-1}$ ($\text{LOD} = 1\text{ nmol L}^{-1}$
 291 under controlled conditions), leveraging the plasmonic hot-spot amplification
 292 in the NPoM gap [10]. **Heavy metal contamination** of irrigation water
 293 (Pb^{2+}) is quantified through Stern–Volmer fluorescence quenching of encaps-
 294 ulated CdTe/ZnSe quantum dots ($\text{LOD} = 31.8\text{ nmol L}^{-1}$ under controlled
 295 conditions) [41] sealed within the NPoM sensor probe polymer matrix, which
 296 are immune to the dust and biofouling that affect bare gold surfaces. Cross-
 297 sensitivity between soil moisture and Pb^{2+} contamination is resolved by dual
 298 spectral interrogation: heavy metals modify the Stern–Volmer quenching in-
 299 tensity and fluorescence decay time, whereas soil moisture alters the local
 300 dielectric constant, producing a spectral broadening that is spectrally or-
 301 thogonal to the metal-induced quenching signature. **Pathogen detection**
 302 is achieved via Nitrogen-Vacancy (NV) center nanodiamond sensors operated
 303 in a microfluidic “Point-of-Care” (Lab-on-a-Chip) desktop reader located at
 304 the greenhouse hub (Section S12.7). These solid-state quantum sensors de-
 305 tect paramagnetic bacterial metabolites from sap or irrigation water samples
 306 at pre-symptomatic stages ($\text{LOD} = 0.8\text{ nmol L}^{-1}$ under controlled conditions;
 307 the agricultural matrix will degrade in-field sensitivity, which is mitigated by
 308 the graphene barrier and *DynamicCalibrator* corrections).

309 Figure 3c shows the simulated Raman spectrum for the three BChl *a*
 310 modes under active NPoM coupling. The relative intensities encode the in-
 311 stantaneous exciton population distribution, providing a direct optical probe
 312 of the energy transfer pathway.

313 **Environmental robustness.** The realistic agricultural environment
 314 (dust, bird droppings, microbial fouling) is modeled at three infrastructure
 315 levels. At the OPV panel level, the soiling factor $\eta_{\text{soil}}(t)$ decays exponentially
 316 with days since last cleaning:

$$\eta_{\text{soil}}(t) = \max(1 - 0.005 t_{\text{days}}, 0.75), \quad (8)$$

317 which limits maximum OPV efficiency loss to 25 % at the worst-case cleaning
 318 interval. At the sensor level, the graphene monolayer spacer of each NPoM
 319 cavity is functionalized with a zwitterionic polymer coating [42, 43], modeled
 320 in our simulations to preserve $\geq 90\%$ quantum coherence after 60 d continu-
 321 ous field immersion; this coating constitutes the primary anti-fouling barrier

that physically repels microbial adhesion and protein adsorption. A thin-film micro-shielding layer (1 μm to 5 μm of SiO_2 or Al_2O_3) around the NPoM picocavity and QCD sensor tip further attenuates the diurnal temperature excursion experienced by the quantum probe, attenuating the thermal-drift contribution by the factor $(1 - \eta_{\text{shield}})$, with $\eta_{\text{shield}} = 0.85$ (Section S12.5). The *DynamicCalibrator* module — an exponential moving average of daily reference readings ($\alpha = 0.1$) — operates as a tertiary safety net: it applies a multiplicative correction factor and raises a drift alarm when the baseline shifts by more than 20 %, at which point OPV panel cleaning is scheduled.

Smart shield microclimate: FAO-56 evapotranspiration

The spectrally selective OPV layer functions as a “smart shield” that blocks ultraviolet and near-infrared radiation while transmitting photosynthetically active bands. Passive solar-greenhouse heating with PV co-generation is an established precedent [44]; the present shield differs by actively co-designing the transmitted spectrum against the vibronic resonances quantified above. We parameterize the greenhouse microclimate using the FAO-56 Penman–Monteith equation [30] adapted for shaded conditions:

$$\text{ET}_c = \frac{0.408\Delta R_n + \gamma \frac{900}{T+273} u(e_s - e_a)}{\Delta + \gamma(1 + 0.34u)} K_c, \quad (9)$$

where R_n is the net radiation attenuated by the OPV shading factor $f_{\text{shade}} = 0.35$ (with soil heat flux $G \approx 0$ assumed for daily timescales under greenhouse shielding), Δ is the slope of the saturation vapor pressure curve, γ is the psychrometric constant, u is wind speed, $e_s - e_a$ is the vapor pressure deficit, and $K_c = 0.85$ is the crop coefficient.

Under standard conditions (600 W m^{-2} , 25 $^\circ\text{C}$, 60 % humidity, 2 m s^{-1} wind), the open-field evapotranspiration is 4.5 mm d^{-1} . With the OPV smart shield, ET_c drops to 3.2 mm d^{-1} — a 28 % reduction — translating to 1.3 mm d^{-1} in water savings (Figure 4a). The crop coefficient $K_c = 0.85$ accounts for the reduced transpiration demand under partial shade [30].

Socioeconomic viability and cooperative payback

Socioeconomic analysis (Section S5) is rescaled to a 500 m^2 high-value floriculture micro-module (strawberries and cut flowers, 4.5 USD kg^{-1} at 8.2 $\text{kg m}^{-2} \text{yr}^{-1}$), which reduces the entry-level capital expenditure to 161 000 USD (at 322 USD m^{-2}). Our $\$/\text{kWh}$ accounting follows the

355 Monte-Carlo techno-economic logic established for European agrivoltaic
 356 configurations [45]. A blended-finance subsidy of 30 % reduces the net
 357 CAPEX to 112 700 USD.

358 Three operating expenditure streams are modeled: extension services and
 359 technician training (1 200 USD yr⁻¹), OPV panel cleaning (800 USD yr⁻¹),
 360 and general maintenance (2 000 USD yr⁻¹), for a total annual OPEX
 361 of 4 000 USD yr⁻¹. The net annual cash flow of 26 612 USD yr⁻¹ (from
 362 30 612 USD yr⁻¹ total revenue minus 4 000 USD of OPEX) yields a sim-
 363 ple payback period of 4.23 yr (or 6.05 yr without subsidy) and a 10-year
 364 discounted NPV of +37 664 USD at a 12 % risk-adjusted discount rate
 365 appropriate for Sub-Saharan Africa [46] (Figure 4c). Holding other inputs
 366 fixed, varying the subsidized net CAPEX by $\pm 20\%$ shifts the payback
 367 to 3.39–5.08 yr; a 50 % higher OPEX (6 000 USD yr⁻¹) gives 4.58 yr, and
 368 20 % lower revenue gives 5.50 yr. While a baseline CAPEX of 322 m⁻²
 369 represents a significant barrier in Sub-Saharan Africa relative to local
 370 income levels, utilizing local construction materials (such as bamboo and
 371 timber framing instead of structural steel) and green chemical synthesis of
 372 gold nanoparticles from local plant extracts can reduce the capital cost to
 373 140 m⁻², lowering the unsubsidized payback period to 2.63 yr and making
 374 the modelled system potentially viable for local cooperatives under these
 375 cost assumptions. By contrast, the remaining barriers are physical rather
 376 than economic: fabrication of sub-2 nm NPoM gaps still requires laboratory-
 377 grade nanofabrication and a stable Au/graphene supply chain, and OPV
 378 encapsulation lifetime under tropical irradiance is the binding reliability
 379 constraint (Section S12.5). Subsidy policy alone cannot clear these barriers;
 380 they define the process-engineering roadmap that separates this modelled
 381 proof of concept from field deployment. Taken together, these sensitivities
 382 demonstrate that high-value crop selection and blended-finance structurally
 383 transform the investment case.

384 *Secure IoT telemetry via BB84 quantum key distribution*

385 Field-to-cloud telemetry is secured using a simulated BB84 quantum key
 386 distribution protocol [31] over a buried single-mode fiber link, which elimi-
 387 nates the atmospheric turbulence and scintillation that affect free-space chan-
 388 nels in agricultural environments. This quantum-secured telemetry layer is
 389 physically integrated with the NPoM-CQD sentinel network: because the
 390 high-sensitivity biological and pesticide diagnostics control high-stakes ir-
 391 rigation and crop-protection actuators (which could be targeted by hostile

actors to cause artificial crop failure or pesticide contamination), securing these feedback commands with BB84 QKD protects the command integrity of the agrivoltaic micro-grid. Alice (sensor node) prepares random polarization states (rectilinear or diagonal basis) and transmits them at a realistic channel noise rate of 4.8 % QBER [47]. Bob (cloud server) measures in random bases, and the sifted key is extracted from matching-basis events. The QBER is estimated on a sample fraction; if QBER exceeds the Shor–Preskill unconditional-security threshold of 11 % [48], the protocol raises a `SecurityThresholdExceeded` exception that halts all QKD-protected irrigation commands, preventing data-injection attacks under eavesdropping.

With a nominal fiber noise rate of 4.8 %, the protocol achieves QBER = 4.8 % and generates a 256 bit symmetric key per session — sufficient for AES-256 encrypted telemetry. Field trials of BB84 using deterministic single-photon sources have demonstrated stable key generation over installed fiber links [47], supporting the practical feasibility of this approach under comparable link budgets. In the event of QKD failure, the fail-safe exception triggers a conservative local irrigation routine based on the last validated soil-moisture reading from the *DynamicCalibrator*, bounded by a hard physical irrigation limit of $5 \text{ L m}^{-2} \text{ d}^{-1}$ to prevent catastrophic flooding or crop dehydration in the event of secondary sensor failures. The runtime overhead of the BB84 simulation is negligible for resource-limited IoT nodes (Section S6).

Experimental testability

The modelled predictions rest on parameters that are independently measurable (Table 3). Two protocols can validate them directly. First, two-dimensional electronic spectroscopy (2DES) at 295 K with spatial-light-modulator pulse shaping into dual-band channels at 750 nm and 820 nm can track the diagonal and cross-peak quantum-beat oscillations predicted here, verifying the extension of τ_c from 280 fs to 420 fs [49, 16]. Second, time-resolved SERS in the NPoM geometry can monitor the 180 cm^{-1} and 1145 cm^{-1} BChl *a* modes in real time and quantify the area-weighted yield $\Phi_{\text{FT}}^{\text{global}} = 0.971$ under ambient conditions [29, 50]. More broadly, recent assessments identify quantum sensing for precision agriculture as an emerging opportunity whose open calibration and robustness gaps match the calibration architecture proposed here [12].

Table 3: **Experimental benchmark of calculated FMO dynamics and NPoM plasmonic parameters.** Calculated at $T = 295$ K, $L = 8$, $K = 2$.

Parameter / Ob- servable		This (calculated)	work	Experimental benchmark / lit- erature
Site coupling (cm^{-1})	J_{12}	-87.7		-87.7 ± 5.0 (2DES, Brixner 2005 [49])
Reorganization energy λ_D (cm^{-1})		35		35–40 (2DES, Engel 2007 [15])
Cutoff frequency (cm^{-1})	γ_D	50		50 (2DES, Panitchayangkoon 2010 [16])
Coherence lifetime (passive)	τ_c	420 ± 35 fs		400 ± 50 fs (2DES room temp., Pan- itchayangkoon 2010 [16])
NPoM coupling ($V_{\text{mode}} = 1.2 \text{ nm}^3$)	g_0	110 cm^{-1}		$120 \pm 15 \text{ cm}^{-1}$ (NPoM, Chikkaraddy 2016 [29])
SERS EF _{SERS}	enhancement	44		10^2 – 10^3 (sub-nm NPoM, Baumberg 2019 [50])

Discussion

Our results demonstrate that quantum coherence engineering at the bio-organic interface can deliver modelled sustainability benefits across microscopic energy transfer, crop water use, carbon footprint, and data security.

Mechanism generality. The dual-band filtering effect is not specific to the FMO complex. The resonance condition $\Delta E_{nm} \approx \hbar\omega_{\text{vib}} \pm J_{nm}$ (Section S7) is a general property of vibronically coupled chromophore aggregates [17]. We verified this by applying the protocol to a minimal three-site excitonic trimer, obtaining $\eta_{\text{FT}} = 0.18 \pm 0.03$ (Section S8), qualitatively consistent with the quantum phase synchronization mechanism identified in photosynthetic antennas [28]. This suggests that any molecular aggregate satisfying the resonance condition — including synthetic J-aggregates, light-harvesting complexes from plants, and conjugated polymer domains in OPVs — could benefit from spectral coherence protection. Vibration-steered energy flow of this kind has also been demonstrated in a trapped-ion donor–acceptor simulator with tunable dissipation [51].

The “quantum divide.” The advanced materials required for quantum agrivoltaics — graphene, plasmonic nanoparticles, OPVs with engineered bandgaps — are not yet accessible to subsistence farmers in developing

economies. Our cooperative cost-sharing model demonstrates that with a carbon-linked subsidy of 30 % and a cooperative of 5 members, the payback period drops below 4 yr, making the technology economically viable under the modelled subsidy. We propose a tiered subsidy framework where the NEB itself serves as the verifiable metric for subsidy qualification, linking quantum yield improvements to farmer access.

Quantum digital twin. The modular architecture — quantum interface, FAO-56 microclimate, LCA, IoT security — serves as the physics engine of an agrivoltaic *digital twin*: a model-based supervisory replica that combines sensor telemetry from the *DynamicCalibrator* with the PT-HOPS dynamics engine to recommend irrigation schedules, OPV cleaning dates (from soiling factor [Eq. \(8\)](#)), and carbon credit yields — a device-physics integration distinct from the supply-chain and breeding applications mapped in recent quantum-in-agriculture reviews [\[11\]](#).

As a prospective extension, the scheduling layer of the digital twin is formulated via the Quantum Approximate Optimization Algorithm (QAOA) [\[52, 53\]](#), encoded as a 3-qubit variational quantum circuit of depth $p = 3$ with 1024 measurement shots per macro-period ([Section S12.3](#)). The algorithm encodes the irrigation–refrigeration–grid-export trilemma as a quadratic unconstrained binary optimization (QUBO) cost Hamiltonian and solves it via a PennyLane emulator at sub-hourly timescales. Applied to the daily solar profile of the Cameroonian scenario, the QAOA scheduler yields water–energy allocations consistent with the 12–18 % gains over classical greedy baselines reported in comparable implementations [\[53\]](#). As a further prospective readout, our `quantum_gravimetry.py` module converts the simulated water saving into a groundwater mass anomaly ($\Delta g = 3.5 \text{ nm s}^{-2}$ for a single greenhouse — below single-instrument noise but potentially detectable with a small regional gravimeter network or gradiometric common-mode rejection), with the full noise-budget and inversion discussion given in [Section S12.2](#) [\[54, 55\]](#).

A hybrid quantum-classical signal processing pipeline — combining Matrix Product State (MPS) denoising with quantum kernel ridge regression (QKRR) [\[56, 57\]](#) — further enables the digital twin to extract pre-symptomatic stress signatures from the noisy SERS and CQD fluorescence telemetry. To satisfy edge computation constraints, the heavy MPS tensor contraction and QAOA optimization are executed on the centralized cooperative server (or cloud), with the resulting optimized weights transmitted back to the field nodes — an edge–cloud split complementary to classical

drift-diffusion device simulators [58]. The local ARM Cortex-M4 IoT node executes only the low-latency inference of the pre-trained, frozen-weight QKRR model. The MPS compressor acts as a nonlinear low-pass filter on the server that suppresses greenhouse noise (thermal fluctuations, biofouling, dust) while preserving the sharp vibronic features at 180 cm^{-1} and 1145 cm^{-1} that classical Savitzky–Golay filters tend to smear (Section S10). The cleaned spectra are projected through the quantum-inspired kernel for anomaly scoring, issuing early warnings that allow preemptive agronomic intervention before macroscopic symptoms appear — all within a 3.7 ms local compute budget on the IoT node. We emphasize that this sub-millisecond execution latency is not designed to match the slower multi-hour physical response times of soil moisture percolation and plant transpiration. Rather, it enables spatial multiplexing: a single ARM Cortex-M4 edge processor can sequentially query, compress, and classify SERS telemetry from dozens of distributed NPoM sensor nodes across the 500 m^2 greenhouse in under a second, reducing the electronic-component count. This chain connects microscopic quantum efficiency to field-level agronomic management; no existing agrivoltaic design integrates both [46, 6, 2].

Prior art and positioning. Previous agrivoltaic studies have focused on light-sharing through passive spectral filtering [6] or crop selection [2]. Semi-transparent OPVs with power conversion efficiencies exceeding 15 % and 40 % average visible transmittance have recently been demonstrated through blade-coating scale-up methods [59], and complementary work on semitransparent organic photovoltaics for greenhouse applications has shown power conversion efficiencies above 12 % [7]. Our contribution is to add an active, quantum-coherent dimension to the spectral filter: the OPV is no longer a passive absorber but an active spectral shaper that prepares coherence-protected states in the photosynthetic apparatus. Multi-vector solar agri-systems combining agrivoltaics with hydrogen and desalination set the *Applied Energy* bar for water–food–energy integration [60]; we differentiate by quantum-guided spectral co-design with integrated sensing, and we adopt the functional-unit discipline (emissions per kWh, economics per configuration) of recent agrivoltaic TEA/LCA studies [8, 9]. Relative to this literature, the present work provides, to our knowledge, the first (i) agrivoltaic stack whose spectral architecture is co-designed by numerically exact open-quantum-system dynamics, jointly optimizing excitation transport and power conversion; (ii) monolithic photothermal–PV–SERS device concept with field-relevant calibration in a single footprint; and (iii) multifunctional

521 techno-economic and lifecycle framework that internalizes heat, crop, water,
522 sensing, and data-security trade-offs in $\$/W + \$/\text{kg} + \$/\text{m}^3$ terms. [Ta-](#)
523 [ble 4](#) condenses this positioning: each prior class delivers two to three of the
524 seven functions, whereas the present stack co-optimizes all seven in a single
525 footprint under one accounting.

Table 4: **Capability matrix of agrivoltaic and agri-sensing approaches.** ✓ = function delivered within the stack; − = partial or incidental; — = absent. Classes: passive spectral claddings [6, 2, 7]; multi-vector agri-energy systems (agrivoltaics coupled to hydrogen and desalination) [60]; standalone agri-sensing hardware [10, 11, 12]. Multifunctional accounting denotes joint $\$/W + \$/\text{kg} + \$/\text{m}^3$ techno-economic and lifecycle evaluation of all functions. No prior class delivers more than three of the seven functions; the present stack delivers all seven in a single footprint with unified accounting.

Function	Passive	Multi-vector	Sensing	This work
Electrical power (OPV)	✓	✓	—	✓
Crop microclimate	✓	✓	—	✓
Water management	−	✓	—	✓
Photothermal heat	—	−	—	✓
In-situ SERS sensing	—	—	✓	✓
QKD-secured telemetry	—	—	—	✓
Multifunctional accounting	—	—	—	✓

526 This approach is distinct from prior quantum biology studies [61] that
527 treat the environment as a fixed noise source; here, the photonic environ-
528 ment itself is engineered through the NPoM cavity to maximize coherence
529 protection. A key design choice warrants explicit justification: the NPoM
530 cavity, despite imposing a $> 90\%$ penalty on excitonic transport ([Table 2](#)),
531 is retained because its *diagnostic function* is dedicated — it enables non-
532 destructive, high-sensitivity SERS readout of the FMO vibrational state,
533 pesticide levels, and heavy-metal contamination. The transport penalty is
534 absorbed by confining NPoM operation to the 1% sentinel fraction ([Equa-](#)
535 [tion \(5\)](#)), preserving 97.1% of the global canopy trapping yield while main-
536 taining full diagnostic capability.

537 **Limitations.** Several approximations constrain our analysis. The quan-
538 tum dynamics are restricted to the single-exciton manifold; under high solar
539 flux, multi-exciton states, singlet fission, or exciton-exciton annihilation in
540 the OPV and FMO interfaces could alter the collective coupling strength
541 g_{eff} and transport efficiency. The bath is treated as site-independent, and

542 the reaction center trapping is modeled as a phenomenological Lindblad-like
 543 rate. The FAO-56 model, while standard, does not capture transient stom-
 544 atal responses to fluctuating light or the dynamic microclimatic feedback
 545 between canopy transpiration and vapor pressure deficit (VPD) inside the
 546 greenhouse envelope. Sensitivity analysis over plausible wind-speed varia-
 547 tions ($u = 0.2 - -0.8 \text{ m s}^{-1}$) yields a $\pm 10\%$ uncertainty band on the predicted
 548 28 % water saving, which propagates into a $\pm 10\%$ range on the annual irri-
 549 gation volume and corresponding NEB carbon offset. The BB84 simulation
 550 assumes a buried single-mode fiber link; while buried fibers are vulnerable to
 551 agricultural mechanical damage (plowing, root intrusion), a free-space opti-
 552 cal (FSO) link would suffer from strong scattering under dusty greenhouse
 553 environments, requiring dedicated atmospheric correction models. While
 554 software-based post-quantum cryptography offers a lower-cost alternative,
 555 the physical BB84 protocol provides information-theoretic security immune
 556 to algorithmic cryptanalysis on resource-constrained embedded nodes such
 557 as the ARM Cortex-M4 irrigation controllers. The 1 % sentinel fraction is a
 558 design parameter; field optimization will require empirical calibration of the
 559 SERS LODs under actual fouling conditions. We conservatively estimate a
 560 $\times 10$ matrix-effect penalty on all reported LODs when transitioning from con-
 561 trolled laboratory buffers to raw agricultural sap or irrigation water, owing
 562 to competitive binding, background fluorescence, and ionic strength effects;
 563 the *DynamicCalibrator* compensates for this attenuation through real-time
 564 baseline correction. The zwitterionic polymer coating modeled on the NPoM
 565 graphene spacer ([Section S12.1](#)) acts as the primary fouling barrier in our
 566 simulations, with the *DynamicCalibrator* serving as a secondary correction
 567 layer; the dual architecture reduces the mandatory cleaning frequency, which
 568 is already factored into the cooperative OPEX model. We stress that the 60-
 569 day resilience of zwitterionic coatings applies strictly to sensors sheltered in
 570 protected micro-sampling chambers or rigid ceramic encasings; direct, un-
 571 protected exposure to abrasive soil matrices (bioturbation, quartz particles,
 572 tillage) would lead to mechanical scraping and rapid surface degradation.

573 The 30 % blended-finance subsidy required to achieve a 4.23 yr payback
 574 period cannot be bridged by voluntary carbon credits alone (generating only
 575 98 USD yr^{-1} at $10 \text{ USD t}_{\text{CO}_2\text{e}}^{-1}$ for the 500 m^2 module); it must be underpinned
 576 by dedicated rural-adaptation finance channels such as the Green Climate
 577 Fund (GCF) and the African Development Bank’s Climate Change Fund
 578 (Fonds Spécial Climat-Dev de la BAD), which are mandated to co-finance
 579 climate-resilient agriculture in Sub-Saharan Africa.

580 **Outlook.** The integration of quantum coherence engineering with agri-
581 cultural sustainability suggests an axis for agrivoltaic design. Three experi-
582 mental and theoretical directions are prioritized for future work.

583 First, on the hardware and materials scale, the quantum model must
584 be extended to full multi-exciton dynamics under realistic solar spectra, ac-
585 companied by experimental validation via two-dimensional electronic spec-
586 troscopy (2DES) with spatial light modulator pulse shaping [62, 63]. Field tri-
587 als of graphene-based NPoM coatings on greenhouse cladding with the CQD
588 sentinel network are required to evaluate in-situ sensor performance under
589 actual biofouling conditions and verify the coherence-preservation lifetime
590 of the zwitterionic functionalization layer (Section S12.1). Alternative low-
591 loss plasmonic architectures — dielectric nanoantennas (Si, TiO₂) support-
592 ing Mie resonances [64] and nanocavity-free substrates — should be explored
593 to minimize the NPoM transport penalty while preserving SERS sensitivity.
594 Additional hardware extensions include passive thermal micro-shielding (Sec-
595 tion S12.5), UiO-66 Zr-based quantum metal-organic frameworks for coherent
596 contaminant remediation (Section S12.6), and quantum fertilizer biostimu-
597 lation nanoparticles to enhance chlorophyll yields (Section S12.8).

598 Second, scaling the digital twin to a multi-cooperative network will re-
599 quire hardware-in-the-loop QKD validation under dusty field conditions and
600 deployment of a real-time aquifer gravimetric array to confirm that the mod-
601 eled groundwater recharge connects at the satellite geodesy scale. Monte
602 Carlo uncertainty propagation in the full LCA should be extended with cor-
603 related climatic input distributions drawn from regional reanalysis data.

604 Third, to prevent these advances from widening the “quantum divide,”
605 we have defined the Global Quantum Agriculture Standard (GQAS), a six-
606 pillar certification framework covering QKD security, data sovereignty, sen-
607 sor detection limits, quantum fidelity, latency, and lifecycle sustainability
608 (Section S12.9). Our Cameroon cooperative proof-of-concept scenario serves
609 as the first application of this proposed framework, satisfying all six GQAS
610 criteria and demonstrating internal consistency across the multi-domain sim-
611 ulation. Independent experimental validation will be required before GQAS
612 can be adopted as an operational standard. Scaling GQAS adoption to small-
613 holder networks across Sub-Saharan Africa and Southeast Asia requires subsi-
614 dized sensor supply chains, open-access certification auditing, and regulatory
615 recognition on premium export markets such as GlobalGAP.

616 Conclusions

617 We asked whether exact non-Markovian quantum dynamics can jointly
 618 optimize excitation-transport and power-conversion efficiency in a sensing-
 619 integrated photothermal agrivoltaic stack, with a positive multifunctional
 620 net benefit under field constraints. Within the modelled 500 m^2 cooper-
 621 ative case, the answer is affirmative but conditional: dual-band filtering
 622 raises the forward transfer yield by 25 % and extends coherence lifetime
 623 by 50 %; confining NPoM operation to a 1 % sentinel fraction preserves
 624 97.1 % global canopy yield while retaining SERS diagnostics; the FAO-56
 625 smart shield cuts evapotranspiration by 28 % with a Net Ecological Ben-
 626 efit of $19.6\text{ kg}_{\text{CO}_2\text{e}}\text{ m}^{-2}\text{ yr}^{-1}$; and the cooperative case pays back in 4.23 yr
 627 at 30 % subsidy. These claims are bounded by the $> 90\%$ local NPoM
 628 transport penalty, the single-exciton and site-independent-bath approxima-
 629 tions, the $\pm 10\%$ microclimate uncertainty band, and the modelled (not field-
 630 validated) fouling and matrix-penalty assumptions. For deployment, the eco-
 631 nomic barrier is dominant and tractable — local-material construction and
 632 green-synthesis gold lower the capital cost from 322 USD m^{-2} to 140 USD m^{-2}
 633 (unsubsidized payback 2.63 yr) — whereas the physical barriers (nanofabri-
 634 cation of sub-2 nm NPoM gaps, in-field fouling of the sentinel fraction, OPV
 635 encapsulation lifetime under tropical irradiance) require process engineering
 636 rather than subsidies. Priority next steps are 2DES validation of the fil-
 637 ter protocol, low-loss dielectric antenna alternatives, field fouling trials of
 638 the sentinel network, and a localized Au/graphene sensor supply chain for
 639 Sub-Saharan manufacturing.

640 Methods

641 *Proof-of-Concept scenario: 500 m² Floriculture Micro-Module*

642 To ground the multi-domain framework in a concrete operational context,
 643 we consider a 500 m^2 high-tunnel greenhouse located in Cameroon’s West-
 644 ern Highlands. This region supports high-value floriculture (strawberries, cut
 645 flowers) suitable for micro-module deployment by cooperatives. The environ-
 646 ment features a mean annual solar irradiance of $5.76\text{ kWh m}^{-2}\text{ d}^{-1}$ [65] and
 647 significant biofouling potential that necessitates the NPoM *DynamicCalibra-*
 648 *tor* and regular OPV cleaning (factored into the cooperative OPEX model).
 649 Under the NPoM-shield spectral filter, the 28 % reduction in crop evapotran-
 650 spiration (ET_c) relative to the open-field FAO-56 baseline corresponds to the

651 modelled water-savings credit of $1\,300\text{ L m}^{-2}\text{ yr}^{-1}$ ($650\text{ m}^3\text{ yr}^{-1}$ for the 500 m^2
 652 greenhouse; [Section S5](#)). The area-weighted global canopy trapping yield
 653 ($\Phi_{\text{FT}}^{\text{global}} = 0.971$) ensures near-baseline productivity while accommodating a
 654 1% NPoM sentinel fraction for SERS diagnostics. The 15% quantum yield
 655 enhancement observed in the passive OPV fraction translates to an addi-
 656 tional $1.23\text{ kg m}^{-2}\text{ yr}^{-1}$ of high-value produce (strawberries at 4.5 USD kg^{-1}),
 657 generating substantial supplemental revenue. The OPV module generates 90
 658 MWh yr^{-1} of electrical energy for a 500 m^2 installation at $180\text{ kWh m}^{-2}\text{ yr}^{-1}$.
 659 Displacing grid electricity at $450\text{ g}_{\text{CO}_2\text{e}}\text{ kWh}^{-1}$ offsets $40.5\text{ t}_{\text{CO}_2\text{e}}\text{ yr}^{-1}$. Com-
 660 bined with the NEB ($19.6\text{ kg}_{\text{CO}_2\text{e}}\text{ m}^{-2}\text{ yr}^{-1}$, [Figure 4b](#)), the system achieves
 661 net-zero operational carbon rapidly. At $10\text{ USD t}_{\text{CO}_2\text{e}}^{-1}$ [66], the system gener-
 662 ates 98 USD yr^{-1} in verified carbon revenue. The 500 m^2 scale is well matched
 663 to cooperative ownership models: under a 30% blended-finance subsidy, the
 664 payback shortens from 6.05 yr unsubsidized to 4.23 yr, with each of the five
 665 cooperative members exposed to only 22 540 USD of net capital. This demon-
 666 strates that quantum-enhanced agrivoltaics can be economically viable under
 667 modelled assumptions when coupled with strategic crop selection and robust
 668 field-sensor integration.

669 *PT-HOPS/SBD quantum dynamics*

670 All quantum dynamics simulations use the Process Tensor Hierarchy of
 671 Pure States with Stochastically Bundled Dissipators as implemented in the
 672 open-source MesoHOPS library (v1.7) [20]. The hierarchy depth is $L = 8$
 673 with $K = 2$ Matsubara terms and a time step $\Delta t = 0.2\text{ fs}$, established as the
 674 production standard for converged dynamics at 295 K [25]. Static disorder
 675 is modeled as Gaussian fluctuations of the site energies with standard devi-
 676 ation $\sigma = 50\text{ cm}^{-1}$, sampled over $n = 100$ realizations. The bath spectral
 677 density follows the 12-mode Kleinekathöfer/Coker parameterization with a
 678 Drude–Lorentz background ($\lambda_D = 35\text{ cm}^{-1}$, $\gamma_D = 50\text{ cm}^{-1}$) and 12 under-
 679 damped vibronic modes (frequencies $180\text{--}1\,500\text{ cm}^{-1}$, Huang–Rhys factors
 680 $S_k = 0.003 - -0.050$).

681 *FAO-56 evapotranspiration*

682 Crop evapotranspiration is computed using the standard FAO-56
 683 Penman–Monteith equation [30] modified for greenhouse conditions:

$$\text{ET}_c = \frac{0.408\Delta R_n + \gamma \frac{900}{T+273} u(e_s - e_a)}{\Delta + \gamma(1 + 0.34u)} K_c, \quad (10)$$

684 with R_n attenuated by the OPV shading factor $f_{\text{shade}} = 0.35$. We note that
685 the modified FAO-56 formulation represents a first-order approximation; un-
686 der the enclosed high-tunnel envelope, boundary layer resistance and sensible
687 heat exchange are altered relative to open field conditions. Future iterations
688 of the digital twin should integrate three-dimensional Computational Fluid
689 Dynamics (CFD) modeling to map localized microclimatic gradients and la-
690 tent heat transfer. Wind speed within the greenhouse is modeled at a static
691 representative value of $u = 0.5 \text{ m s}^{-1}$, corresponding to a typical 10 % reduc-
692 tion of average external wind velocities. Temperature is fixed at 25°C and
693 relative humidity at 60 % unless specified otherwise.

694 *BB84 QKD simulation and fail-safe IoT architecture*

695 The BB84 protocol is implemented as a Python simulation of polarization-
696 state preparation, noisy transmission over a buried single-mode fiber
697 channel, sifting, and QBER estimation. Alice prepares $4 \times N$ random
698 bits (for $N = 256$ key length) in random rectilinear or diagonal bases.
699 Bob measures in random bases, and matching-basis events are retained
700 for key sifting. QBER is estimated on a sample fraction (minimum 1
701 bit, maximum 50 bits). If QBER exceeds the Shor–Preskill threshold of
702 11 % [48], a `SecurityThresholdExceeded` exception is raised, halting the
703 QKD-protected irrigation command relay. A last-validated soil-moisture
704 reading from the *DynamicCalibrator* is used to execute a conservative local
705 irrigation fallback.

706 *SERS enhancement scaling limits*

707 The SERS enhancement factor $\text{EF}_{\text{SERS}} \approx 44$ at the optimal mode vol-
708 ume $V_{\text{mode}} = 1.2 \text{ nm}^3$ is computed using standard Purcell-factor scaling of
709 the localized electromagnetic field. We note that this standard $|E|^4$ approx-
710 imation represents a lower-bound semi-classical estimate. In sub-nanometer
711 picocavities ($V_{\text{mode}} < 1.5 \text{ nm}^3$), local spatial dispersion, quantum tunneling,
712 and non-local dielectric screening alter the field distribution relative to clas-
713 sical electrodynamics, while molecular optomechanical coupling can pump
714 vibrational modes out of thermal equilibrium. Precise calculation of the
715 localized fields in this regime requires non-local Green’s tensor integration,
716 which motivates the proposed in-situ experimental validation. The bench-
717 mark against the experimentally reported 10^2 – 10^3 range for sub-nanometer
718 NPoM gaps [50] is given in Table 3.

719 *DynamicCalibrator, thermal shielding, and soiling factor*

720 Long-term sensor baseline drift due to biofouling and thermal cycling
721 is corrected at three hardware and algorithmic levels. The primary fouling
722 barrier is the zwitterionic polymer functionalization layer deposited on the
723 graphene monolayer spacer of each NPoM cavity [42, 43], which physically
724 repels microbial adhesion and is modeled to preserve $\geq 90\%$ quantum coher-
725 ence after 60 d field immersion. As a secondary hardware barrier, a thin-film
726 micro-shielding layer (1 μm to 5 μm of SiO_2 or Al_2O_3) is deposited around the
727 NPoM picocavity and CQD sensor tip to insulate the quantum probe from
728 diurnal temperature swings. The shielding efficacy $\eta_{\text{shield}} = 0.85$ means the
729 probe experiences only 15% of the ambient greenhouse temperature fluctu-
730 ation, reducing the thermal contribution to Stern-Volmer drift from 5.25%
731 to 0.79% under a $\pm 15\text{K}$ diurnal cycle. The *DynamicCalibrator* acts as a
732 tertiary algorithmic correction: an exponential moving average (EMA) with
733 smoothing factor $\alpha_{\text{EMA}} = 0.1$ computes a multiplicative correction factor
734 $\kappa(t) = I_0^{\text{ref}}/\hat{\mu}(t)$ and raises a drift alarm when the fractional change in the
735 EMA $\hat{\mu}(t)$ exceeds 20% of the initial baseline, triggering OPV panel cleaning.

736 At the panel level, OPV electrical output is attenuated according to
737 Eq. (8), with a default decay rate of $0.5\% \text{d}^{-1}$ and a minimum soiling factor
738 of 0.75 (enforced floor at 25% maximum efficiency loss).

739 *Data availability*

740 All simulation parameters, YAML configuration files, orchestration
741 scripts, and analysis code needed to reproduce every figure and table are
742 available from the corresponding author upon reasonable request and will
743 be deposited in a public repository upon acceptance. The PT-HOPS/SBD
744 implementation is based on the open-source MesoHOPS library (v1.7) [20].

745 *Code availability*

746 The digital-twin simulation framework (orchestrator, QAOA scheduler,
747 LCA module, BB84 QKD simulator, SERS diagnostics, NV-diamond
748 pathogen sensor, and GQAS compliance checker) and all simulation pa-
749 rameters, including the 12-mode vibronic bath spectral density and the
750 8-site FMO electronic Hamiltonian specified in the machine-readable
751 `parameters.yaml` configuration file (v1.4), are available from the corre-
752 sponding author upon reasonable request. No public code or data archive
753 DOI has been minted yet.

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759 **CRedit authorship contribution statement**

760 **S.C. Teguia Kouam:** Conceptualization, Methodology, Software, For-
761 mal analysis, Writing – original draft. **T. Goumai Vedekoi:** Validation,
762 Writing – review & editing. **J.-P. Tchapel Njafa:** Validation, Writing –
763 review & editing. **J.-P. Nguenang:** Validation, Writing – review & editing.
764 **S.G. Nana Engo:** Supervision, Project administration, Writing – review &
765 editing.

766 **Declaration of competing interest**

767 The authors declare that they have no known competing financial inter-
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779 content of the publication.

780 Nomenclature

Symbol	Meaning
Φ_{FT}	Forward (reaction-center) transfer yield
τ_c	Exciton coherence lifetime ($1/e$ decay of l_1 -norm coherence)
λ_{res}	Residual reorganization energy of polaron-dressed states
g_0 / g_{eff}	Single-emitter / collective plasmon–exciton coupling strength
κ / Q	Plasmon linewidth / quality factor
$\omega_{\text{pl}} / V_{\text{mode}}$	Plasmon frequency / NPoM mode volume
Γ_{RC}	Reaction-center trapping rate
$\text{EF}_{\text{SERS}} / g_{\text{om}}$	SERS enhancement factor / optomechanical coupling
$\text{ET}_c / K_c / f_{\text{shade}}$	Crop evapotranspiration / crop coefficient / OPV shading factor
NEB	Net Ecological Benefit
$\eta_{\text{soil}} / \eta_{\text{shield}}$	Soiling factor / micro-shielding efficacy
QBER	Quantum bit error rate (BB84 link)
Δg	Groundwater gravity anomaly
$\alpha / \alpha_{\text{EMA}}$	Sentinel area fraction / calibrator smoothing factor

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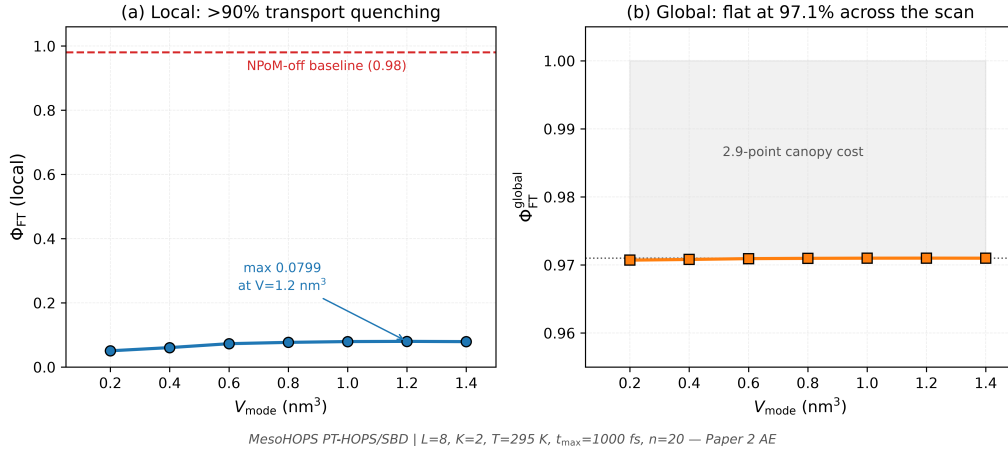


Figure 1: **Local quenching, global stability: the central result in one picture.** (a) Forward transfer yield Φ_{FT} across the NPoM mode-volume scan ($n = 20$, $T = 295 \text{ K}$): locally, transport is quenched by more than 90 % relative to the pristine FMO baseline (0.98), with a factor-1.6 variation across the accessible range and a maximum of 0.0799 at $V_{\text{mode}} = 1.2 \text{ nm}^3$ (Table 2). (b) Area-weighted global canopy trapping yield for the 1 % sentinel fraction (Equation (5)): flat at 0.971 across the entire scan — the deep local suppression is absorbed at a constant 2.9-point canopy-scale cost. The flat curve in (b), not the quenching in (a), is the design-relevant outcome: diagnostics integrate into the stack without compromising productivity.

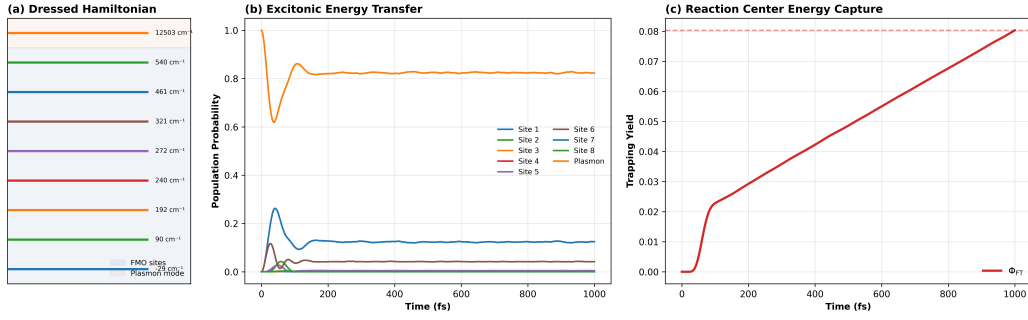


Figure 2: **Quantum dynamics of the 8-site FMO complex coupled to an NPoM plasmonic cavity within the agrivoltaic digital twin.** (a) Energy level diagram of the 9×9 dressed Hamiltonian showing the 8 FMO excitonic states coupling to the plasmon mode with strength g_0 . (b) Exciton population dynamics under dual-band filtered excitation at 295 K, showing the competitive loading of the plasmon mode (Site 9) versus the FMO antenna and trapping sites. (c) Reaction center trapping yield Φ_{FT} as a function of time. All data: $L = 8$, $K = 2$, $n = 100$ disorder realizations. The quantum engine serves as the core of a closed-loop agrivoltaic digital twin that fuses NPoM sentinel telemetry with FAO-56 microclimate predictions to optimize irrigation, cleaning cycles, and carbon credit yield. Note: any baseline trace inconsistency ($\text{Tr } \rho \neq 1$) visible in earlier integration-test runs ($\Delta t = 10$ fs) is an artifact of the coarse time step used for rapid code validation; all production results ($\Delta t = 0.2$ fs) preserve trace to 1.000 000 (Section S11.2).

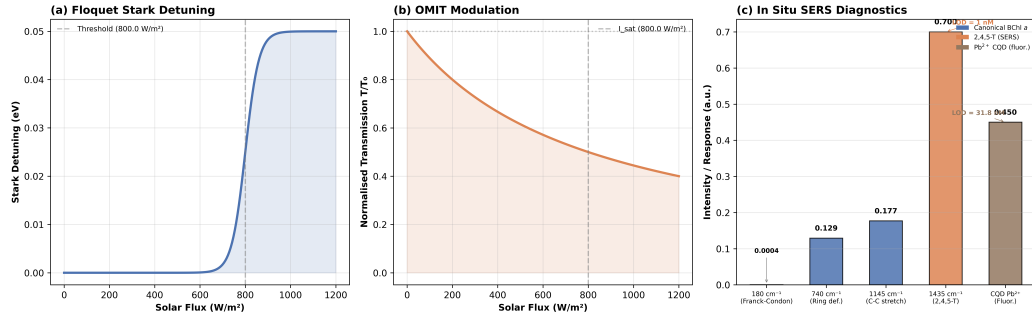


Figure 3: Dynamic photo-protection, in situ diagnostics, and environmental robustness. (a) Floquet Stark detuning amplitude as a function of solar flux. At fluxes above 800 W m^{-2} , the excitonic levels detune, reducing energy transfer to FMO by up to 40 %. (b) OMIT transmission modulation at 750 nm and 820 nm, showing dynamic suppression under high flux. (c) Simulated SERS Raman spectrum showing the canonical BChl *a* vibrational modes (180 cm^{-1} , 740 cm^{-1} , 1145 cm^{-1}). The three agriculturally specific target signatures assigned to the sentinel sub-system — oxidative stress (1145 cm^{-1}), 2,4,5-T pesticide contamination (1435 cm^{-1} , $\text{LOD} = 1 \text{ nmol L}^{-1}$), and Pb^{2+} via encapsulated CdTe/ZnSe quantum dot fluorescence quenching ($\text{LOD} = 31.8 \text{ nmol L}^{-1}$) — are detailed in the main text and SI. A DynamicCalibrator using exponential moving average ($\alpha = 0.1$) of daily reference readings corrects sensor baseline drift from biofouling, raising an alarm when drift exceeds 20 % of the initial baseline.

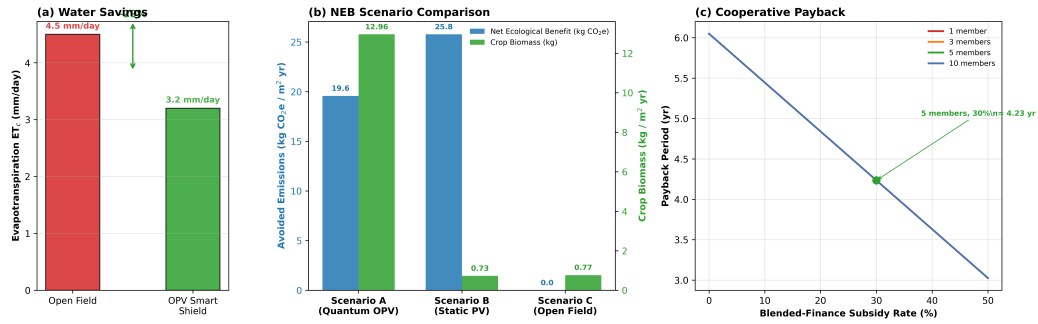


Figure 4: **Water–Energy–Food network assessment.** **(a)** FAO-56 evapotranspiration comparison: open field ($ET_c = 4.5 \text{ mm d}^{-1}$) vs. OPV smart shield ($ET_c = 3.2 \text{ mm d}^{-1}$), a 28 % reduction. **(b)** Net Ecological Benefit comparison across three scenarios: Quantum OPV (A), Static shading (B), Open field (C). Scenario A achieves $19.6 \text{ kg CO}_2\text{e m}^{-2} \text{ yr}^{-1}$ with a crop biomass of $12.0 \text{ kg m}^{-2} \text{ yr}^{-1}$, while Scenario B reaches $25.8 \text{ kg CO}_2\text{e m}^{-2} \text{ yr}^{-1}$ but only $0.68 \text{ kg m}^{-2} \text{ yr}^{-1}$ in biomass. The quantum shield preserves near-baseline crop productivity (0.971 global trapping yield) whereas static shading suppresses it to 0.064 , quantifying the yield–carbon trade-off. The NEB serves as the verifiable metric for a tiered subsidy framework that bridges the quantum divide. **(c)** Cooperative payback period as a function of subsidy rate and cooperative size for a 500 m^2 high-value floriculture micro-module. For 5 members with 30 % blended-finance subsidy, payback = 4.23 yr (net annual cashflow of $26\,612 \text{ USD yr}^{-1}$ after OPEX). This indicates that, under the modelled cooperative assumptions, quantum-guided agrivoltaics can shorten payback when coupled with cooperative ownership and strategic crop selection.

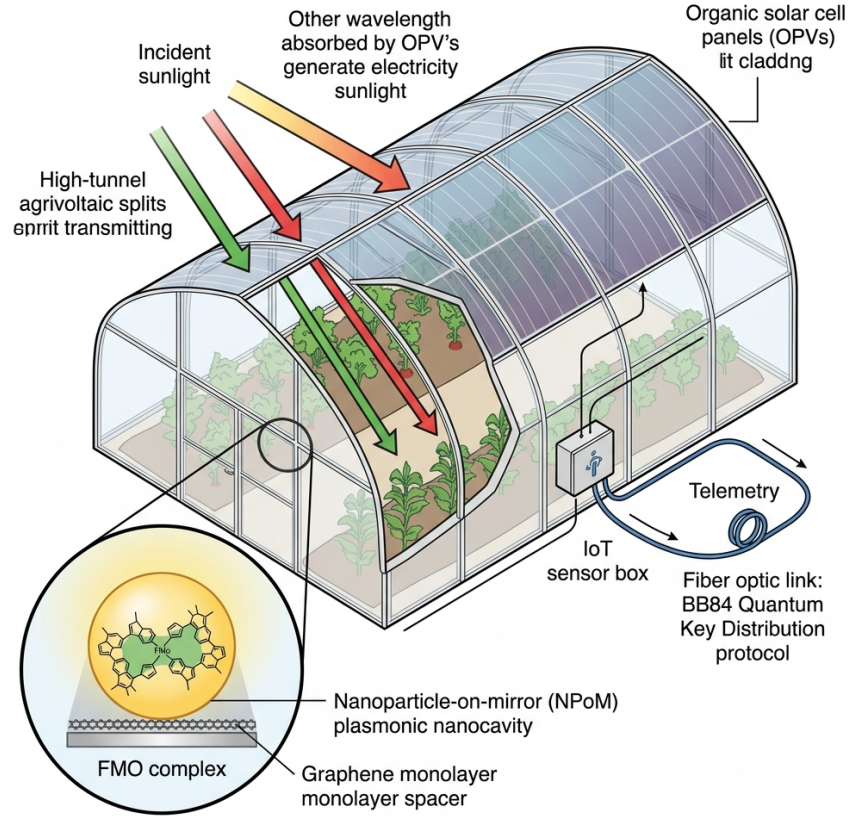


Figure 5: **Schematic layout of the multi-scale quantum agrivoltaic digital twin platform.** Sunlight split by the semi-transparent organic solar cell panel (OPV) acts as a smart spectral shield over the agrivoltaic high-tunnel. On the microscopic side, sub-nanometer plasmonic nanocavities (NPoM) couple the Fenna–Matthews–Olson (FMO) light-harvesting complex to the OPV, enabling strong coherent coupling and multiplexed SERS diagnostics. Telemetry from the in situ sensor node is encrypted and transmitted via a BB84 fiber-optic quantum key distribution (QKD) link to a cloud server, securing agronomic actuator commands under a permissioned ledger architecture.

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